The Cyclotomic Factor Cascade as a Supercritical Branching Process

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Abstract

We model the factor chain induced by the third cyclotomic polynomial $\Phi_3(p) = p^2+p+1$ as a Galton–Watson branching process on the set of odd primes. For each prime p in a cascade seeded by a finite set S_0, we count the number of *new* odd prime factors of $\Phi_3(p)$ not already present in the running set S. Over 300 seed configurations generating 12,927 parent–child observations, we find a global mean offspring rate $\bar{\mu} = 1.1747$, placing the process in the supercritical regime. We observe a phase transition: $\mu(p) < 1$ for primes $p \lesssim 200$ and $\mu(p) > 1$ for $p \gtrsim 200$, consistent with Stewart's theorem that $P^+(\Phi_3(p)) > p^{2-\varepsilon}$ for large p. The offspring rate increases further for higher cyclotomic polynomials ($\mu \approx 2.99$ for Φ_5, $\mu \approx 1.71$ for Φ_7). These measurements provide a quantitative explanation for the empirical observation that no finite set of odd primes is closed under the σ-induced factor map, offering probabilistic evidence against the existence of odd perfect numbers.

1 Introduction

A natural number N is *perfect* if $\sigma(N) = 2N$, where σ denotes the sum-of-divisors function. By the Euclid–Euler theorem, every even perfect number has the form $2^{p-1}(2^p - 1)$ with $2^p - 1$ prime. Whether any odd perfect number (OPN) exists has been open since antiquity.

Euler proved that any OPN must have the form

$$N = q^\alpha \cdot p_1^{2e_1} \cdots p_k^{2e_k}, \tag{1}$$

where $q \equiv \alpha \equiv 1 \pmod 4$ is the *special prime* and all $e_i \geq 1$. Modern results impose increasingly severe necessary conditions: $N > 10^{2200}$ [7], $k \geq 10$ [5], and largest prime factor exceeding 10^8 [3].

A key technique in these results is the *factor chain* (or *network method*): since $\sigma(N) = 2N$ and σ is multiplicative, every odd prime dividing $\sigma(p_i^{2e_i})$ must itself divide N, hence belong to $S = \{q, p_1, \ldots, p_k\}$. This *closure condition* propagates: new primes force new factors of σ, generating a cascade. Nielsen [4] and Ochem–Rao [7] exploit this cascade computationally within branching search trees to derive lower bounds.

In the simplest case $e_i = 1$, the relevant factor is $\sigma(p^2) = \Phi_3(p) = p^2 + p + 1$, where Φ_3 is the third cyclotomic polynomial. The cascade then takes the form:

$$S_{n+1} = S_n \cup \big\{\text{odd prime factors of } \Phi_3(p) : p \in S_n\big\}.$$

If this process stabilizes at a finite set, closure is achieved. If it diverges, no finite closed set exists.

Our contribution. We reinterpret this cascade as a *Galton–Watson branching process* and compute the mean offspring rate $\mu(p)$ empirically. To our knowledge, this probabilistic framing—specifically, the computation of μ as a function of prime size and the identification of a subcritical-to-supercritical phase transition—has not appeared in the literature. While the divergence of factor chains is well known empirically, the branching process framework provides a quantitative explanation grounded in classical probability theory: a Galton–Watson process with $\mu > 1$ survives (diverges) with positive probability, and the extinction probability decreases exponentially with population size [1].

2 Setup and definitions

Definition 1 (Cyclotomic cascade)**.** Let S_0 be a finite set of odd primes. Define the Φ_3-*cascade* by

$$S_{n+1} = S_n \cup \big\{q \text{ odd prime} : q \mid \Phi_3(p) \text{ for some } p \in S_n\big\}.$$

Write $S_\infty = \bigcup_{n \geq 0} S_n$. The cascade *closes* if S_∞ is finite.

Definition 2 (Offspring count)**.** For a prime p encountered at round n of the cascade, define its *offspring* as the set of odd primes dividing $\Phi_3(p)$ that are not already in S_n at the time p is processed:

$$\mathcal{C}(p, S_n) = \big\{q \text{ odd prime} : q \mid \Phi_3(p),\ q \notin S_n\big\}.$$

The offspring count is $X_p = |\mathcal{C}(p, S_n)|$.

In a classical Galton–Watson process, each individual produces offspring independently with a common distribution. Our setting departs from this idealization in two ways: (1) the offspring distribution depends on the arithmetic properties of p (it is not identically distributed), and (2) offspring are suppressed if they already appear in S_n (a depletion effect). Nevertheless, the mean offspring rate μ retains its classical significance: if $\mu > 1$, the population grows exponentially in expectation.

Proposition 3 (Residue restriction [4, Theorem 2])**.** *If q is an odd prime dividing $\Phi_n(p)$, then either $q \mid n$ or $q \equiv 1 \pmod{n}$. In particular, odd prime factors of $\Phi_3(p)$ satisfy $q = 3$ or $q \equiv 1 \pmod{3}$.*

3 Computational method

We generated three families of seed sets:

- 200 singletons $\{p\}$ for primes $p \equiv 1 \pmod{3}$, $7 \leq p < 10{,}000$;
- 60 pairs $\{3, p\}$ for such primes p;
- 40 triples $\{3, p_1, p_2\}$ from small primes $\equiv 1 \pmod{3}$.

For each of the resulting 300 seeds, we iterated the Φ_3-cascade for up to 12 rounds or until $|S| > 200$. Factorization used trial division up to 10^4 combined with deterministic Miller–Rabin primality testing (witnesses $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$, unconditionally correct below 3.3×10^{24}). For each prime p processed, we recorded its offspring count X_p.

Total runtime: 0.5 seconds on an Apple Mac Mini M4, using only the Python standard library. All code is available at [11].

4 Results

4.1 Global offspring rate

Over 300 cascades, we recorded 12,927 parent–child observations:

Quantity	**Value**
Total parent primes	12,927
Total offspring generated	15,187
Mean offspring $\bar{\mu}$	**1.1747**
Cascades closed (S_∞ finite)	0 / 300
Cascades diverged	300 / 300

Table 1: Global statistics for the Φ_3 cascade over 300 seed configurations.

Since $\bar{\mu} > 1$, the cascade is supercritical in the Galton–Watson sense. No cascade closed.

4.2 Offspring distribution

The empirical offspring distribution is concentrated on small values:

Offspring k	**Count**	**Fraction**
0	1,351	0.105
1	8,570	0.663
2	2,497	0.193
3	421	0.033
4	82	0.006
≥ 5	6	< 0.001

Table 2: Offspring distribution. The mode is $k = 1$ (66% of parents produce exactly one new prime).

The distribution resembles a shifted geometric with mean $\mu \approx 1.17$, consistent with a mildly supercritical branching process. The extinction probability for a Galton–Watson process with this offspring distribution, estimated via the generating function $\phi(s) = \sum p_k s^k$, is $\rho \approx 0.73$—high, but strictly less than 1.

4.3 Phase transition: $\mu(p)$ as a function of prime size

Binning the offspring counts by $\lfloor 2 \log_{10} p \rceil / 2$ (bins of width 0.5 in $\log_{10}$), we observe a clear transition:

$\log_{10}(p)$	$\sim p$ range	n	$\mu(p)$	Regime
0.5	3–5	300	0.867	subcritical
1.0	6–17	592	0.840	subcritical
1.5	18–56	1,108	0.956	subcritical
2.0	56–178	2,212	0.895	subcritical
2.5	178–562	1,880	1.138	**supercritical**
3.0	562–1,778	1,802	1.407	**supercritical**
3.5	1,778–5,623	1,524	1.472	**supercritical**
4.0	5,623–17,783	776	1.171	**supercritical**
4.5	17,783–56,234	372	1.911	**supercritical**
5.0	56,234–177,828	470	1.402	**supercritical**

Table 3: Mean offspring $\mu(p)$ by prime size. The transition occurs near $p \approx 200$.

The transition from subcritical to supercritical near $\log_{10}(p) \approx 2.3$ (i.e., $p \approx 200$) is consistent with the following theoretical prediction.

Theorem 4 (Stewart [9]). *For $f(x)$ irreducible of degree $d \geq 2$ over $\mathbb{Z}$,*

$$P^+\big(f(n)\big) > n^{1+1/(d-1)-\varepsilon}$$

for all $n > N_0(\varepsilon)$, where P^+ denotes the largest prime factor.

For Φ_3 ($d = 2$), this gives $P^+(\Phi_3(p)) > p^{2-\varepsilon}$, meaning the number of digits of the largest prime factor roughly doubles each generation. This forces $\mu(p) \geq 1$ for large p, since $\Phi_3(p) \approx p^2$ must have at least one prime factor of size $\gg p$, which is necessarily new.

4.4 Residue class dependence

Consistent with Proposition 3:

$p \bmod 3$	μ	n
$\equiv 0$	0.867	300
$\equiv 1$	1.182	12,627

Table 4: Offspring rate by residue class. Primes $\equiv 2 \pmod 3$ do not appear as factors of Φ_3.

The value $\mu \approx 0.87$ for $p = 3$ reflects the special role of 3 (which divides $\Phi_3(p)$ whenever $p \equiv 1 \pmod 3$, providing no new information). The bulk of the cascade consists of primes $\equiv 1 \pmod 3$, for which $\mu = 1.182 > 1$.

4.5 Largest child ratio and connection to Stewart

For each prime p in the cascade, we computed the ratio $\log_{10}(\text{largest child})/\log_{10}(p)$. Over 300 primes with $p > 10$:

$$\text{mean ratio} = 1.339, \qquad \text{median ratio} = 1.345.$$

Stewart's theorem predicts this ratio tends to 2.0 for large p. The observed value ≈ 1.34 for primes up to $\sim 10^4$ suggests the Stewart regime is not yet fully reached, but the trend is upward, consistent with $\mu(p)$ increasing with p.

4.6 Higher cyclotomic polynomials

For a non-special prime p with exponent $2e$ in an OPN, the relevant factor is $\sigma(p^{2e}) = \prod_{d|(2e+1)} \Phi_d(p)$. The dominant new factor is $\Phi_{2e+1}(p)$ of degree $\varphi(2e+1) \geq 2$ for $e \geq 1$. We measured μ for the first offspring generation of Φ_5 and Φ_7:

	Degree	μ	Stewart exponent
$\Phi_3(p) = p^2 + p + 1$	2	2.987	$1 + 1/1 = 2.0$
$\Phi_5(p) = p^4 + \cdots + 1$	4	2.340	$1 + 1/3 \approx 1.33$
$\Phi_7(p) = p^6 + \cdots + 1$	6	1.707	$1 + 1/5 = 1.2$

Table 5: Mean offspring for higher cyclotomic polynomials (first generation, 150 primes $\equiv 1 \pmod 3$).

All three are supercritical. Since $\varphi(2e+1) \geq 2$ for all $e \geq 1$, Stewart's theorem applies to every non-special exponent. In a realistic OPN with multiple exponents active simultaneously, the effective μ is the sum over all active Φ_n, making the cascade diverge even faster.

5 Implications for odd perfect numbers

The closure condition for an OPN requires $S_\infty = S_0$—the cascade must stabilize immediately. In the branching process framework, this corresponds

to *immediate extinction*: every individual in the first generation must be sterile ($X_p = 0$ for all $p \in S_0$).

For a set S_0 of k primes, the probability of immediate extinction under our empirical offspring distribution is approximately $(p_0)^k = (0.105)^k$, which is $\approx 10^{-10}$ for $k = 10$ (the minimum number of distinct prime factors required by Nielsen [5]). But immediate extinction is necessary *at every round*, not just the first. The probability of the cascade closing within r rounds is bounded by $\rho^{|S_r|}$, where $\rho \approx 0.73$ is the per-individual extinction probability. Since $|S_r|$ grows exponentially when $\mu > 1$, the survival probability approaches 1 exponentially fast.

Remark 5. This is a heuristic argument, not a proof. The Galton–Watson model assumes independence of offspring, which is only approximately true in our setting (the residue restriction of Proposition 3 introduces correlations). A rigorous proof would require showing that the correlations do not reduce μ below 1—equivalently, that the depletion effect (primes already in S) does not dominate the growth. Our data suggest it does not: the depletion effect is strongest for small primes (where $\mu < 1$), but these are outnumbered by large primes (where $\mu > 1$) as the cascade progresses.

Remark 6. Pomerance's heuristic [8] argues against OPNs via the global probability that $\sigma(N) = 2N$ for a random odd number of the appropriate form. Our approach is complementary: we argue against OPNs via the *local* branching rate of the factor cascade. The two perspectives are independent and reinforce each other.

6 Relation to prior work

The factor chain method originates with Sylvester [10], who manually traced small cascades. Nielsen [4] systematized this into the "network method" used to prove $k \geq 9$, later extended to $k \geq 10$ [5]. Ochem and Rao [7] pushed the lower bound on N to 10^{1500} (later 10^{2200}) using branching search trees built on factor chains. Fletcher, Nielsen, and Ochem [2] combined sieve methods with factor chains.

Our contribution is not the cascade itself, which is well established, but the probabilistic framing: (1) computing the empirical offspring distribution and mean μ across 12,927 observations; (2) identifying the subcritical-to-supercritical phase transition near $p \approx 200$; (3) connecting $\mu(p) > 1$ to Stewart's theorem as the theoretical explanation; (4) measuring μ for Φ_5 and Φ_7, showing universality of supercriticality.

The BYU group's work on spoof perfect factorizations [6] proved that infinitely many spoofs exist, implying that approaches using only the multiplicative structure of σ cannot resolve the OPN problem. Our results are consistent with this: the branching process model captures the multiplicative structure and shows it is generically divergent, but cannot distinguish primes from composites (the essential difference between real OPNs and spoofs).

7 Open questions

1. **Rigorous supercriticality.** Can one prove that $\mu(p) > 1$ for all primes $p > p_0$, without relying on the empirical distribution? Stewart's theorem guarantees at least one large child, but bounding the *expected number* of new primes requires controlling the smooth part of $\Phi_3(p)$ as well.

2. **Correlation structure.** The offspring counts are not independent across primes in the same cascade. Quantifying the correlation decay—or proving it is negligible—would make the branching process model rigorous.

3. **Higher exponents in combination.** In a realistic OPN, multiple exponents are active simultaneously. The effective branching rate is the sum over all active Φ_n. A joint analysis incorporating the constraint $\sum e_i \geq C(k)$ from known bounds could yield stronger results.

4. **Connection to abc.** The conjecture abc implies effective versions of Stewart's theorem. Does abc give a computable N_0 that closes the gap between the supercritical regime ($p > 200$ empirically) and the theoretical threshold?

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